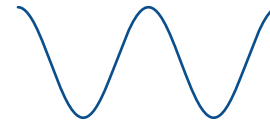
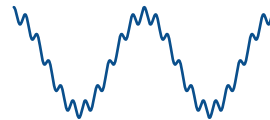
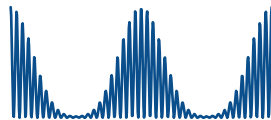
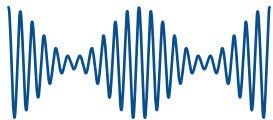


The cortical column as an AM radio receiver



AM field $s(t)$

carrier f_c
envelope Ω



**Nonlinear
detector**

firing-rate sigmoid
 $\sigma(v)$, curvature σ''



**Tuned
resonator**

2nd-order synapses
near Hopf, f_0



**Demodulated
output @ Ω**

alpha rhythm
recovered

RF signal

*square-law /
diode detector*

*RLC tank
(tuned)*

audio (message)